

Enhanced AA & AP
Unit 4A Test Review

Graph one period of each function and complete the information for each equation.

1. $y = \cos\left(3x - \frac{\pi}{2}\right)$ amp. = _____ period = _____ phase shift: _____ vertical shift: _____

2. $y = 3 \sin\left(x - \frac{\pi}{3}\right) - 2$ amp = _____ period = _____ phase shift: _____ vertical shift: _____

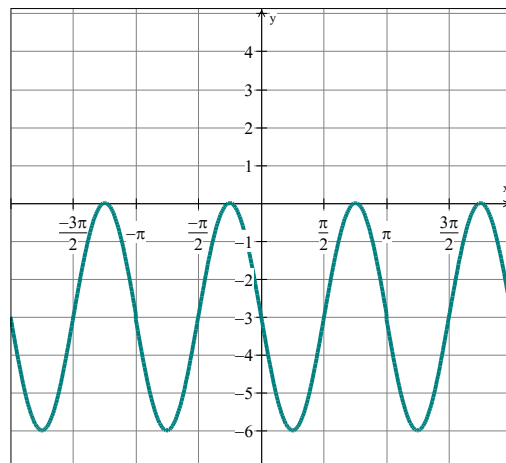
3. $y = -4 \cos\left(\frac{\theta}{2} + \pi\right) - 6$ a = _____ period = _____ phase shift: _____ vertical shift: _____

4. Find $\cos\theta$ the given that $\cot\theta = -\frac{6}{5}$ and $\sec < 0$.

5. Write an equation for each of the curves using the graph at the right.

Sine _____

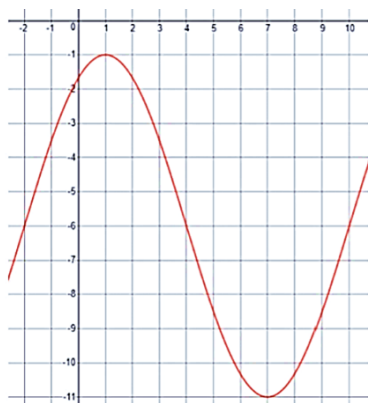
Cosine _____



6. Write a sine equation using the table

x	-5π	-3π	$-\pi$	π
y	2	-1	2	5

7. Write a cosine equation using the graph



Write a sine or cosine equation for the problems below.

8. One particular July 4th in Galveston, TX, high tide occurred at 9:36 a.m. At that time the water at the end of the 61st Street Pier was 2.7 meters deep. Low tide occurred at 3:48 p.m., at which time the water was only 2.1 meters deep. Assume that the depth of the water is a sinusoidal function of time with a period of half a lunar day (about 12 hours 24 minutes).

9. An earthquake occurred at 9:40 a.m. on Nov. 1, 1755, at Lisbon, Portugal, and started a tsunami in the ocean. It produced waves that traveled more than 540 ft/sec (370 mph) and reached a maximum height of 60 ft. The period of the waves was 30 minutes. Write an equation for the height of the wave as a function of time. Assume midnight is $t=0$.

10. Write an equation of a sine curve with amplitude = 3, period = 5, phase shift = $\frac{\pi}{4}$, vertical shift = up 2

11. Write an equation of a cosine curve with amplitude = $\frac{1}{4}$, period = $\frac{\pi}{2}$, phase shift = $\frac{\pi}{2}$, vertical shift = 0. Also, describe the transformations from the parent function $f(x) = \sin(x)$.

12. The minute hand of a clock is 6 inches long. How far does the tip of the minute hand move in 25 minutes. Leave answer in exact form.

13. A water sprinkler sprays water over a distance of 30 feet while rotating through an angle of 135° . What area of the lawn receives water? Leave in exact form.

14. Evaluate without a calculator.

$$\cos\left(\frac{2\pi}{3}\right)$$

$$\sec\left(-\frac{5\pi}{4}\right)$$

$$\sin\left(\frac{15\pi}{2}\right)$$

$$\cot\left(\frac{7\pi}{6}\right)$$

$$\csc\left(\frac{\pi}{2}\right)$$

$$\cos\left(-\frac{5\pi}{3}\right)$$

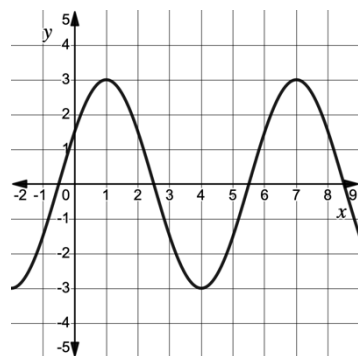
$$\sin\left(\frac{3\pi}{4}\right)$$

$$\tan 7\pi$$

$$\tan\left(-\frac{4\pi}{3}\right)$$

15. A portion of the graph of a periodic function $y = f(x)$ is shown. On each of the following intervals, determine if f is increasing or decreasing and if the graph of f is concave up or concave down.

Interval	Increasing or Decreasing?	Concave up or Concave down?
$22 < x < 23.5$		
$13 < x < 14.5$		
$38.5 < x < 40$		



16. The average monthly temperature in Denver, Colorado is recorded in the table below:

a. Using your calculator, find the sinusoidal regression function, $T(m)$. You will use the SinReg option. You will not need to change the default settings – Iterations: 3, Period: [Leave blank]. Round to 100th.

Month (m)	Temperature (Degrees Fahrenheit)
January	29.7
February	33.4
March	39
April	48.2
May	57.2
June	66.9
July	73.5
August	71.4
September	62.3
October	51.4
November	39
December	31

b. What is the midline of your function? What does this value represent in context of the data?

c. Find the period of your function? Does this number make sense in context of the data? Explain.

d. Use your Regression function to estimate the Temperature in Denver for May 6th of next year.